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Factors Affecting School Dropout and Completion in Mexico: Does Agency Matter?

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ABSTRACT *This paper examines which factors influence the opportunities of completing upper-secondary education in Mexico, rather than dropping out of it. Drawing on Sen's capability approach and Bourdieu's sociocultural-reproduction perspective, two research hypotheses are formulated to provide possible explanations of persons' decisions to reach higher levels of schooling. These hypotheses are tested using data from Mexico's First Survey of School Dropouts, carried out in 2011. The results of this study indicate that the probabilities of completing upper-secondary education are associated with several factors: socioeconomic and demographic variables, the type of upper-secondary institution attended, human agency and educational experiences. The evidence presented in this study supports research hypotheses based on Sen's and Bourdieu's approaches. Regarding the hypothesis from the capability approach, the study data show that human agency is not only intrinsically valuable but also instrumentally important, for reaching higher educational levels. The findings of this study also bear out Bourdieu's argument that cultural and economic capital contribute to explain why students make progress in school, although, this paper challenges Bourdieu's view of the role of freedom of educational choice. Finally, some implications for educational policies are discussed in the last section of this paper.*

KEYWORDS: Education, Agency, School dropout, Cultural capital, Quantitative methods

Introduction

Education continues to be a key element for fostering human development. However, many persons do not enrol at compulsory education or face difficulties while attending school and, therefore, leave education without developing key skills and abilities and without obtaining a formal educational certificate. For several reasons, the issue of dropping out of school has been recognized as a serious public concern. For instance, research has shown that school dropouts face disadvantages in the labour market; they have lower probabilities of being employed and lower chances of getting better-paid jobs than school completers (Rumberger 2011; Brekke 2014; INEE 2018). In addition, Thornberry, Moore, and Christenson (1985) show that dropping out is associated to arrest after controlling for sociodemographic characteristics such as age, race, and social status. Henry, Knight, and

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Thornberry (2012) also note that earlier school disengagement is associated to dropout as well as several problem behaviours such as subsequent involvement in delinquency and problem substance use. Rumberger (2011) highlights that the odds of having access to health care and of participating in further education and training to work are lower for high-school dropouts than for persons with higher levels of schooling. Thus, there is evidence that dropping out of school may have a broad range of individual, economic and social consequences.

Mexico achieved almost universal participation in primary education and expanded participation in secondary and post-secondary schooling. However, only 6 out of ten individuals between the ages of 15 and 17 attend upper-secondary classes, even though upper-secondary education has been compulsory since 2012 (National Institute for the Evaluation of Education [Spanish acronym: INEE], 2018), added to which approximately 15 out of every 100 people who enrol in upper-secondary education drop out, usually during their first year of studies (INEE 2018). These facts have made it more necessary to examine the factors influencing dropping out of upper-secondary education as it is intended in this article.

The central question that will be addressed in this article is: to what extent do young people's sociocultural and ascriptive characteristics, economic resources, institutional factors, human agency and educational experiences influence the chances of completing upper-secondary education instead of dropping out? In order to address this question, this paper has been divided into five main parts. The first part presents a literature review on school dropout and completion; the second part presents the research hypotheses; the third section shows data and methods used for carrying out this study; the fourth section discusses the empirical findings. Finally, some conclusions and implications for educational policies are discussed in the last section.

Previous Studies on Secondary-School Dropout and Completion

The literature on school dropout is diverse and, in general, it has reported that this issue is usually influenced by more than one factor. For instance, the international comparative study of the predictors of school dropout and completion in 13 different countries carried out by Lamb and Markussen (2011) highlights the importance of family background, demographic factors, individual characteristics, experiences of school; educational context and community and economic settings as important predictors of dropping out.

Other studies have examined the influence of school-related factors on dropout rates. Rumberger and Thomas (2000) show that dropout-rate variations for a sample of high schools in the USA can be attributed to differences among the backgrounds of the students in question, as well as other factors such as school resources, school structure and school processes. Another study, carried out by Rumberger (1995), found that the issue of dropping out from high schools is partially explained by several structural features of the said schools, as well as by the atmosphere in them. Research by Roderick (1994) into the influence of level repetition on high-school dropout rates found that repetition of a school year is an important predictor of dropout levels in high schools in Massachusetts.

With regard to Latin America, Bassi, Buso, and Muñoz (2014) explored trends in graduation and dropout rates over time in 18 Latin American countries, determining that secondary-school graduation rates increased between 1990 and 2010, whereas dropout rates decreased during the same period. They noted that the period when the probability of school dropout was highest moved from the transition between primary and secondary education to a later stage of secondary schooling, showing that students are spending increasing amounts of time in school.

In Mexico, analysis of the phenomenon of dropout has relied, to some extent, on censuses and household surveys such as the National Housing and Population Census and the National Survey of Occupation and Employment. Studies carried out using national surveys have shown that students drop out of upper-secondary education mainly because they find their schools irrelevant to their lives and interests (Székely 2010). Other research SEP-SEMS (2012) indicates that dropping out of school is associated with family-related factors, socioeconomic circumstances and previous educational experiences. Although these studies have offered important insights into dropouts from upper-secondary education at the national level, they limit themselves to identifying a broader range of influences on the said phenomenon.

Other research projects have examined specific Mexican states. For instance, that carried out by Blanco (2014) explores interruptions in school attendance at different educational levels in Mexico City. Abril et al. (2008) explore the reasons for dropping out of high school in the state of Sonora, finding that the main reasons for abandoning school were socioeconomic circumstances, poor academic achievement, academic failure and lack of interest in studying.

Overall, most of the literature on school dropout stresses that dropout rates are usually influenced by numerous factors such as social, individual and school variables. However, less attention has been paid to analysing the role play by the type of institution attended in upper secondary education on school dropout using survey methodology, as I intend to do in this paper. In addition, little effort has been made to examine the phenomenon of school dropout from a capability perspective.

Furthermore, those who have carried out research in the specific field of education and capabilities note that much more research is needed to better understand how human agency is related to education. For instance, Flores-Crespo (2002, 538) states that: "Explaining the relationship between inequality and education from a functionalist viewpoint runs the risk of failing to discuss how individuals act within social structures." In a similar vein, Malhotra (as cited in Brighouse and Untherhalter 2010) notes that "macro-level studies are especially weak in measuring agency and often do not use a relevant conceptual framework." Thus, few studies that have been carried out in the area of education and capabilities using quantitative methodologies to measure agency and capabilities, which might be partially explained by the lack of data from surveys and large-scale assessments based on a capability approach. This paper explores the role of agency on educational attainment using survey data from Mexico.

Research Hypotheses

Drawing on the capability approach and the sociocultural-reproduction theory, two research hypotheses were formulated and will be tested. The strategy—further described below—that has been adopted in order to ascertain the explanatory power of these hypotheses consists in determining the type and strength of the association between their key explanatory factors and the outcome variable. The first research hypothesis to be tested in this paper stems from the capability approach and asserts that:

> Individuals' ability to move up through the different levels of upper-secondary and tertiary education is influenced by human-agency factors such as the degree of freedom/power to make choices about education-related matters. If persons have more freedom/power to make educational decisions, such as choosing the type of institution they wish to study in, they are more likely to attain higher levels of education.

In order to test this hypothesis, we will explore the extent to which the explanatory variable called "degree of freedom/power to choose school" is associated with increased probabilities of completing upper-secondary education. More specifically, if young people have a certain degree of freedom to participate in making decisions about the school they will attend, and if this freedom has a positive impact on their likelihood of successfully completing upper-secondary education, then this hypothesis will be accepted. Conversely, if authorities dictating which school a young person will attend, makes it more probable that s/he will finish upper-secondary school, then the capability-based hypothesis will be rejected.

The second research hypothesis, which explores the explanatory power of some of the assumptions arising from Bourdieu's sociocultural-reproduction theory, is as follows:

> Individuals' decisions to move up through the different levels of upper-secondary and tertiary education are influenced by structural factors such as cultural capital and socioeconomic resources. Individuals from more advantageous socioeconomic backgrounds are more likely to participate in upper-secondary and higher education because the said backgrounds enable them to remain in school longer.

We will employ the parents' education variable to determine levels of cultural capital and use a household-possessions index as a measure of socioeconomic resources. Persons from more advantageous backgrounds are defined as ones who have parents with higher levels of schooling and more material household possessions. The hypothesis stemming from Bourdieu's theory of sociocultural reproduction will deemed to be confirmed if higher educational qualifications and greater amounts of material resources positively correlate with the likelihood of completing upper-secondary education.

Data and Methods

Data comes from the School Dropout Survey (SDS), the sample of which is representative of the number of dropouts in Mexico in 2011. The said results constitute a significant achievement, since gathering information about dropouts is often a complex task due to the challenges inherent in following students' trajectories over the years.

The sample of respondents to the SDS was aged between 14 and 25 when the data were gathered. However, for purposes of this paper, the said sample is restricted to respondents aged between 16 and 25 for three main reasons: first, the probability that people have attended upper-secondary school increases after the age of 16; second, most of the respondents aged 14–15 are still studying lower-secondary education; third, international statistics on dropout from—and completion of—upper-secondary-education are usually calculated for individuals aged 16 or more, and, therefore, focusing on the 16–25 age group would appear to make comparisons between the results for Mexico and those for other countries more feasible.

As it happens with all secondary data, the SDS faces some limitations. For instance, the survey is a cross-sectional study which means that the information was collected in the present to explore some aspects of young people's previous educational and life experiences. According to Blaikie (2010), since retrospective studies collect information about past experiences, there is a chance that respondents do not remember some facts or it might also occur: "a possible reconstruction of the past under the influence of subsequent experiences" (Blaikie 2010, 203). Most data of the SDS should be interpreted retrospectively.

To answer the research question and test the research hypotheses, a set of binomial logistic-regression models are estimated, for all of which the dependent variable is binary, coded "1" if the person completed upper-secondary education, and coded "0" if the person dropped out. The explanatory variables were selected for theoretical reasons and taking into account data availability. It was of special interest to incorporate in the analysis variables that allow operationalising concepts related to the capabilty approach. The School Dropouts Survey offers a broad range of variables (sociodemographic, individual, educational and institutional) which might impact on persons' educational attainment. The key explanatory variables included in the statistical models, and their merits and limitations, are discussed in the next lines.

SocioCultural Resources/Social Origin

For Bourdieu (1997) educational qualifications are a dimension of cultural capital and Breen and Jonsson (2005) note that a great deal of social research commonly operationalizes social origin using the variables parental education or parental occupation. In this study, the variable highest level of parents' education is used as an indicator of individuals' cultural capital and social origin.

To be specific, the variable parental educational is measured by the last educational level that parents attended. The variable comprises three categories: (1) no parents with lower secondary education, (2) at least one parent with upper secondary education and (3) at least one parent studied higher education.

A limitation is that by using a single indicator of social origin, the effects of family background might be underestimated. It is desirable to use multidimensional measurements of sociocultural resources and social origin; however, data on other indicators of these dimensions such as parental occupation were not available in the SDS.

Gender

Numerous studies have recognized that gender is associated to individuals' educational attainment and outcomes. In the survey, gender was self-reported by the individual. This variable is measured as a dummy variable so it comprises two categories: male and female respondents.

Ethnicity

A typical strategy followed in educational and household surveys to identify indigenous persons in Mexico is by using the variable language spoken at home. The survey includes one question that asks whether individuals speak or understand an indigenous language. The question has two possible answer options: (1) speaking an indigenous language and (2) not speaking an indigenous language. For this research, the dummy variable on language spoken at home is used as an indicator of person's ethnic origin. However, this study recognizes that there are some others dimensions of ethnicity such as nationality, skin colour, race and religion.

Prior Academic Performance

Lucas and Beresford (2010, 52), points out that test scores and grades are common measures of academic performance. In Mexico, the criteria to assign grades are quite homogenous among lower-secondary schools but there is more variability on this aspect among

institutions in the upper secondary level. In this study, academic performance is measured by grades in lower secondary education. The variable "grades in lower-secondary" has a scale from six to 10 and six is the minimum requirement to pass lower secondary education. The survey does not provide information about young people's examination results in certain areas of knowledge.

Type of Institution Attended

Previous research has reported that the type of institution attended makes an impact on individuals' educational attainment and outcomes (Kerckhoff 1993; Shavit and Muller 2000). These studies have made a great emphasis on classifying schools by the extent to which they provide academic or vocational education.

The Mexican education system offers different pathways to study upper secondary education including academic, vocational and technical options. However, this study proposes a more refined classification of the upper secondary schools which takes into account not only the distinction between academic and vocational education but also other sources of institutional differentiation such as the prestige or status of schools, administrative ascriptions, funding sources and curricular contents. Thus, the classification of institutions comprises six main categories: (1) academic traditional institutions (high-status); (2) academic open and distance-learning schools (low status); (3) technological institutions managed by the federal government; (4) technological institutions managed mainly by the states governments; (5) technical institutions and (6) private schools.

Educational Aspirations (Functioning)

Drawing on the capability approach, Hart (2013) proposes a conceptualization of aspirations in terms of functionings and capabilities. Hart notes that the functioning of aspiring is different from the capability to aspire. The former is actively constructed by abstract thinking and might further be developed through a variety of external forms of expression. The latter is conceived as the freedom and possibility to aspire. For Hart, if individuals face limitations to freely think and express their aspirations, then their freedom to choose functionings might be affected.

Hart conceptualizes educational aspirations as functioning. More specifically, Hart (2013, 1572) points out:

> an individual may demonstrate the functioning of aspiring by saying that they aspire to go onto higher education but they may have experienced limited capability to aspire to other ways of being and doing they may have reason to value.

Thus, in Hart's accounts, which this study adopts, aspiration to higher education is conceived as a functioning. The data source in which this study relies does not provide information to assess individuals' capability to aspire. However, the variable educational aspirations have a limitation: it merely reflects persons' revealed aspirations and such aspirations might be subject to change over time due to internal and external influences.

Human Agency (Freedom to Choose School)

Sen (1985) defines agency in terms of a person's freedom to do and act in order to reach the goals and values he/she reasonable views as relevant. Drawing on the work of Sen (1985, 1992, 1999) and Rowlands (1997), Ibrahim and Alkire (2007) develop a set of indicators to

measure agency and empowerment as well as a strategy to link these dimensions with survey questions. The indicators include four dimensions: empowerment as control over personal decisions, empowerment as autonomy to make choices, empowerment in community and empowerment as change. For Ibrahim and Alkire (2007, 389), the indicator empowerment as the ability to make choices, is frequently operationalized using data on "who makes the decisions" in some specific sphere of life.

In line with Ibrahim and Alkire, this study conceives an individual's ability to make educational choices as a proxy dimension of empowerment and agency. To be specific, the variable degree of freedom/power to choose school has three categories: it is coded 0 if individuals participate in the selection of school it is coded 1 if parents make the school choice and it is coded 2 if they are assigned to schools by institutional authorities. Thus, the extent to which individuals' participation in educational decision making impacts on educational attainment will be explored.

Ibrahim and Alkire (2007) note that by merely asking who makes a choice, it overlooks that individuals might freely delegate decision-making to another person due to that they are uninterested or lacking time for making such a decision. Thus, it may occur that individuals are not interested on choosing school. Another possible case is that choices among youngsters, parents and institutions coincide and, if this happens, we can assume that such synergy of choices increases the likelihood that students continue in school. Conversely, if there is a conflict of choices (e.g., an individual's choice is not the same that the one made by the institution) we can assume that it would be more complicated for students to remain in school. Unfortunately, the survey in which study relies does not provide further information on the synergy of choices and conflict of choices.

Index of Risk Events

In the school dropouts survey, there are seven personal events which young people may have faced and adversely affect their studies. These are: (1) parental problems, (2) unsafety in the pathway to school, (3) moving from home during studies (4) sickness or death in family, (5) marriage, (6) pregnancy, (7) family preferences towards education (e.g., no support of female education). All of them might increase the risk of dropping out and of no participating in upper secondary education.

From the seven variables described above, an index of risk events will be estimated by summing the "yes" answer option of each question. The affirmative answers will indicate that the risk event indeed occurred during young people's educational experience. All the affirmative values will receive a score of "1" and they will be added together to create a scale from zero to seven. The highest value in the scale will indicate the highest number of risk events that might increase the propensity to not participate or dropout from upper secondary education.

Index of Home Possessions

Opción interesante para agregar al modelo

According to Bourdieu, economic capital is the most material type of "capital" and it can appear as money, properties/material rights or even as titles of nobility (Bourdieu 1997). In this paper, an index of material properties at home will be used as an indicator of economic capital. In the School Dropouts Survey, respondents were asked if they had several possessions in their households. The list of possessions includes 14 goods and services: washing machine, stove, car, microwave, potable water, TV, DVD, refrigerator, cement floor, landline, internet, toilet with drainage and subscription to TV programmes.

Similarly to the elaboration of the index of risk events, an index of home possessions is constructed by adding the affirmative answer of each item. Each "yes" answer is scored 1 and it indicates that the person has the possession at home. The range of this index is from zero to 15. The more possessions individuals have, the higher the value of the index.

Index of School Disengagement

A scale is constructed from a set of binary variables on individuals' attitudes and preferences towards education which indicates a lack of interest or disposition to attend school. These variables are: (1) dislike studying, (2) disagreement with the disciplinary rules of the school, (3) thinking that studying is not very useful, (4) pursuing transferring studies to another school, (5) expulsion from school because of indiscipline, (6) expulsion from school because of failing exams, (7) thinking that having a job is more important than studying, (8) dislike school for being far away from home. This variable is a multidimensional measurement of individuals' preferences towards education and, therefore, it is to some extent linked to the capability approach. Finally, this scale is an ordinal variable that ranges from zero to six.

Urbanization

The variable geographical location comprises 2 categories: it is coded 0 if individuals live in urban areas (above 100 000 inhabitants) and semi-urban areas (between 2500 and 100 000 inhabitants) and it is coded 1 if individuals live in localities with have between 501 and 2,499 inhabitants.

In the following sections, the empirical results that will answer the research question posed in this study, and test the validity of its research hypotheses, are presented and discussed. After presenting the descriptive findings, we go on to present the results of the statistical modelling.

Empirical Results

Descriptive Statistics

The descriptive statistics show variations between the respective groups of dropouts and graduates from upper-secondary education in several dimensions. More specifically, they show the proportions of graduates and dropouts broken down by parents' education, gender, ethnicity, age, type of upper-secondary institution attended, freedom to choose an upper-secondary school, and access to scholarships (Table 1). Moreover, the mean values of the variables: grades, index of household possessions, index of school-disengagement and index of risk-events for graduates are compared with those for dropouts (Table 2).

The first aspect worthy of note is the fact that 81% of the total number of respondents who have at least one parent with higher education complete upper-secondary education, while only 18.8% drop out of it. For students who have parents that entered upper-secondary education, the completion rate is 47%, while the dropout rate is 52%. Table 1 also shows that 62.6% of the total number of female respondents completed upper-secondary education, compared with 57.4% of male ones.

Furthermore, Table 1 shows that 31% of the total number of respondents aged 17 graduated from upper-secondary education, compared with 64% of respondents aged 18 or over. These figures suggest that young people are more likely to complete upper-secondary education after the age of 18, a finding that coincides with those of studies carried out at the

Table 1. Descriptive statistics.

	Dropouts	Graduates	Total
Parents' education			
No parents with upper-secondary education	47.3	52.7	3002
At least one parent with upper-secondary education	37.4	62.6	752
At least one parent with higher education	18.8	81.2	810
Gender			
Male	42.5	57.4	2195
Female	37.3	62.6	2369
Ethnicity			
Non-indigenous	39.7	60.3	4351
Indigenous	40	60	213
Age			
16	85	15	207
17	68.79	31.21	402
18	35.6	64.4	680
19 and older	35.2	64.7	3275
Type of upper-secondary institution attended			
Private	28.7	71.3	406
Academic General institutions (high-status)	36.8	63.2	2176
Open and distance-learning systems	56.3	43.7	182
Federal technological schools	44	56	1015
State-level technological schools	35	65	183
Technical	53	47	479
Other	40.4	59.6	123
Access to scholarships			
No access to scholarships	43.2	53.8	3741
Access to scholarships	26	74	823
Degree of freedom to choose the school			
Institute allocated	61.06	38.94	236
Parents' choice	52.22	44.78	198
Individual's choice	37.77	62.23	4130

Note: These percentages were estimated based on weighted data.

international level, which indicate that, in most education systems, the majority of young people tend to complete upper-secondary programmes between the ages of 17 and 20 (Lamb and Markussen 2011). It should be stressed that the proportion of dropouts is higher (85%) for 16-year-olds which suggests that it is more likely that students abandon their studies during the first or second year of upper secondary school. Data from INEE (2018, 337) is consistent with this argument since one out of four students who enrol in upper secondary level abandon their studies during the first scholar year and this number is lower for later school stages.

Additionally, Table 1 shows that 40% of all indigenous students dropped out of upper-secondary education, compared with 60% who completed it. The figures for non-indigenous students are almost the same—i.e., 39.7% and 60.3%, respectively-, indicating that indigenous and non-indigenous students have practically the same upper-secondary-school completion rates.

Table 1 also shows completion and dropout rates according to the type of upper-secondary institution attended. The first point to note is that more people enrolled in open and distance—learning systems and technical, schools dropped out than graduated. Conversely, significantly more students graduated from private schools, technological schools managed by the state governments, and academic general schools (71%, 65% and 63% respectively) than dropped out of them.

Table 2. Summary statistics for continuous variables.

	Statistics					Comparison between means (*t*-test)			
Variables	Obs.	Mean	Std. dev.	Min	Max	Dropouts	Graduates	*t*-stat	*p*-Value
Grades in lower-secondary education	4,564	8.23	0.75	6	10	7.98	8.47	−23.02	0.00
Household possessions	4,564	10.46	3.07	0	15	9.71	11.15	−16.22	0.00
Disengagement from school	4,564	1.38	1.57	0	12	1.83	0.96	19.44	0.00
Risk events	4,564	0.58	0.88	0	7	0.74	0.42	12.45	0.00

Source: My own elaboration.

On the other hand, 74% of scholarship recipients completed upper-secondary education, as opposed to 26% who dropped out, while 43.2% of the students in upper-secondary education who reported that they had no scholarship dropped out, as opposed to 53.8% who graduated.

Furthermore, the capability approach stresses the importance of free choice for people's wellbeing. For instance, with regard to people's freedom to make educational decisions, such as choosing which school they wish to attend, Hart argues:

> From a capability perspective of well-being, freedom in terms of school or college choice would be able to take account of the freedom an individual has to choose between different institutions as well as the individual's satisfaction with their chosen institution (the latter being a current focus of institutional evaluation). (Hart 2013, 2703–2704)

This study seeks to compare the proportion of dropouts and graduates from upper-secondary education according with people's degree of freedom to choose their upper-secondary school. The results reported in Table 1 show that 62.3% of all the people who mentioned that they were able to participate in choosing their school managed to graduate, while 37.7% dropped out, whereas 61% of the total number of people who stated that they were allocated to an upper-secondary institution without being allowed to choose dropped out, as compared to 38.9 who graduated.

Table 2 shows some summary statistics (means, standard deviations and *t*-tests) for continuous variables such as grades in lower-secondary education, household possessions, disengagement from school and risk events. The results indicate that the variable grades range from 6 to 10, while the mean grade for the group of students who dropped out of secondary education is 7.9, as opposed to 8.4 for the group of students who graduated from it. The *t*-test shows that the mean difference is statistically significant (t-statistic = −22.9 and the associated p-value = 0.00); in other words, the mean difference between the grades of these two populations is not zero.

Moreover, the index of household possessions ranges from 0 to 15, while the mean is 10.4 with a standard deviation of 3.07. The household-possessions mean is 9.7 for dropouts from upper-secondary education and 11.15 for graduates from it, while the mean difference between these two populations is not zero (t-statistic = −16.2 and p-value = 0.00). The disengagement-from-school variable ranges from 0 to 12, while the mean difference between dropouts (1.83) and graduates (0.96) with regard to disengagement from school variant is statistically significant, as is the difference between (0.74) and graduates with regard to risk events (.42) (see Table 2).

The descriptive statistics offer some preliminary insights into the characteristics of the groups of dropouts and graduates from upper-secondary education. Below, we discuss the results of the logistic-regression models.

Factors Influencing Completion of Upper-Secondary-Education

The results of the binomial logistic-regression analysis are shown in Table 2. The set of statistical models seeks to determine the factors associated with the probability of completing upper-secondary education rather than dropping out of it, and to test the validity of two research hypotheses that were formulated drawing on Sen's capability approach and Bourdieu's theory of sociocultural reproduction.

The coefficients are expressed in terms of average marginal effects (AME's) in order to compare the regression coefficients across models. Thus, the parents' education variable is

Table 3. Binomial logistic-regression models for predicting the likelihood of completing upper-secondary education instead of dropping out of it (AMEs).

Variable	Model 1	Model 2	Model 3	Model 4	Model 5
Parents' education (Ref. no parents with upper-secondary education)					
At least one parent with upper-secondary education	.099***	.050**	.047**	.054**	.058***
	(.02)	(.02)	(.02)	(.02)	(.01)
At least one parent with higher education	.285***	.204***	.195***	.179***	.160***
	(.01)	(.02)	(.02)	(.02)	(.02)
Index of household possessions		.026***	.026***	.023***	.021***
		(.002)	(.002)	(.002)	(.002)
Age (Ref. 16)					
17		.163***	.166***	.146***	.125**
		(.04)	(.04)	(.04)	(.04)
18		.470***	.467***	.427***	.378***
		(.03)	(.03)	(.04)	(.04)
19 and older		.477***	.473***	.428***	.400***
		(.03)	(.03)	(.03)	(.03)
Female (Ref. male)		.058***	.057***	.041***	.032**
		(.01)	(.01)	(.01)	(.01)
Indigenous (Ref. non-indigenous)		.080**	.074**	.064*	.031
		(.03)	(.03)	(.03)	(.03)
Type of institution (Ref. academic general)					
Private schools			−.007	−.012	−.005
			(.02)	(.02)	(.02)
Open and distance-learning systems			−.161***	−.143***	−.118***
			(.05)	(.04)	(.04)
Federal technological schools			−.04**	−.05***	−.05***
			(.02)	(.01)	(.01)
Technological schools run by state governments			.030	.016	−.019
			(.03)	(.03)	(.03)
Technical schools			−.010***	−.115***	−.092***
			(.02)	(.02)	(.02)
Other/Unclassified			−.004	−.029	−.014
			(.04)	(.04)	(.04)
Freedom to choose school (Ref. Institution's decision)					
Parents' choice				.062	.032
				(.04)	(.04)
Individual's choice				.164***	.128***
				(.03)	(.03)
Index of disengagement from school				−.067***	−.047***
				(.005)	(.005)
Grades in lower-secondary education					.115***
					(.009)
Access to scholarships					.172***
					(.01)
Index of risk events					−.059***
					(.009)
Number of Obs.	4,564	4,564	4,564	4,564	4,564
***R*2**	.042	.113	.120	.170	.236

Notes: My own elaboration using data from the First Survey of School Dropouts in Mexico, 2011. Data weighted using probability weights. Average marginal effects are reported and linearized standard errors are shown in parenthesis.
Key: * $p < .1$; ** $p < .05$; *** $p < .01$.

included in the first model, while all the other sociodemographic variables are included in the second one. Next, institutional variables form part of the logistic-regression analysis, while the fourth model also includes dispositional and agency-related factors and the last model includes educational-experiences variables (Table 3).

Relacion entre padre que estudia e hijo que estudia

The results of Model 1, which examines the net effect of parents' education on the probability of concluding upper-secondary-education, indicate that people who have a parent with university studies are 28.5% more likely to complete upper-secondary education than ones whose of schooling did not extend beyond lower-secondary education, while those who have at least one parent who attended upper-secondary school are 9.9% more likely to complete uppers-secondary education than the members of the reference group.

Conforme crecen aumenta la relación de educación y posesión de bienes

Hay más probabilidad de estudiar siendo mujer, pero es muy bajo su impacto

The results shown in Model 2, which also explore the effects of all the other socio-demographic variables, such as age, gender, ethnicity and an index of household possessions, indicate a 2.6% increase in the probability of completing upper-secondary education for each unit of increase in the index of household possessions. Moreover, people aged 17 are 16.3% more likely to complete upper-secondary education than the members of the reference group, those aged 18 are 47% more likely to complete upper-secondary education than the members of the reference group, and those aged 19–25 are 47.7% more likely to complete upper-secondary education than the members of the said reference group. Additionally, females are 5.8% more likely to finish upper-secondary than males, and indigenous students are 8% more likely to complete the said school cycle than non-indigenous ones. These results were obtained maintaining average values for all the other variables.

Once sociocultural and economic resources, as well as all the other ascriptive factors, have been accounted for, institutional variables are included in the statistical modelling. Model 3 shows that young people who studied in open and distance-learning systems are 16.1% less likely to complete upper-secondary education than ones who attend general academic institutions. Similarly, those who study in technical schools are 10% less likely to complete upper-secondary education than those who attend general academic (high-status) institutions. Furthermore, people who attend federal technological schools are 4% less likely to complete upper-secondary education than ones who study in the reference group.

Apoyando los postulados de Amartya Sen sobre capacidades y poder de elección

Model 4 explores the effects of the variables freedom to choose school and school-disengagement. As shown in Table 3, people who are able to choose their school are 16.4% more likely to complete upper-secondary education than ones whose school is allocated to them by the educational authorities. Moreover, the probability of completing upper-secondary education decreases by 6.7% for each additional point on the school-disengagement index.

Las becas académicas ayudan a la permanecía escolar

Model 5 controls for educational experiences such as grades in lower-secondary education, access to scholarships and risk events. The results indicate that the probability of completing upper-secondary education increases by 11.5% for each unit increase in grades. Moreover, people who have access to scholarships are 17.2% more likely to conclude upper-secondary education than people who have no such access, in addition to which people's likelihood of not completing upper-secondary education increases by 5.9% for each additional point on the risk-event scale.

It also bears noting that the impact of parents' education is modified by other explanatory variables such as the type of upper-secondary institution, freedom to choose school and educational experience. For instance, in Model 1, people who have a parent with university education are 28.5% more likely to complete upper-secondary school than the members of the reference group, whereas, in the last model, which considers the effects of all the other explanatory variables (Table 3, Column Six), the said likelihood decreases to 16%. Similarly, the influence exerted by ethnicity changes upon inclusion of educational-experiences

variables (grades, access to scholarships and risk events during studies) in the logistic-regression analysis. Thus, in Models 2, 3 and 4, indigenous students are more likely to graduate from upper-secondary school, but, after controlling for academic performance and access to scholarships, there is no statistically significant difference between indigenous and non-indigenous students in this regard. Moreover, these results suggest that, apart from age, the strongest predictor of school completion is access to scholarships.

Finally, Table 3 includes the pseudo-*R*-squared values in order to determine the extent to which the predictive power of the regression analysis changes when successive additional explanatory variables are included. It is worth noticing that for Long and Freese (2014, 3765), there are different alternatives for calculating measures of *R*-square that have some similitude to the *R*-square in the linear regression model. These are known as pseudo-*R*-squares. Stata adopts the McFadden's pseudo *R*-square as the default measure. Thus, the McFadden's pseudo *R*-square is reported in the logistic-regression analysis of the statistical analysis because it compares a model with just the intercept to a model with all parameters (see Table 3). Overall, these results suggest that the predictive power of the model improves after taking into account several explanatory variables that extend beyond sociodemographic factors. The results of the multicollinearity test suggest that there are no high levels of correlations among the predictors that merit further consideration. Specifically, for all the variables, the variance-inflation factor is substantially lower than 10, the mean value of the VIF is very close to 1 (1.10), and the tolerance values of all the predictors are substantially higher than .10.

Summary and Discussion

This study set out to explores the factors influencing the probabilities of completing upper-secondary education rather than dropping out of it. Overall, its results indicate that the likelihood of completing upper-secondary is associated with sociocultural and economic resources as well as with ascriptive characteristics (parents' education, gender, ethnicity, age and household resources), institutional factors (type of upper-secondary institution attended), preferences regarding schooling (disengagement from school); a human agency dimension (freedom to choose school), and educational experiences (grades, access to scholarships and risk events).

The findings show that social background contributes to predict school completion, with more cultural and material resources making it more likely that a student will complete upper-secondary education rather than dropping out of it. Thus, the explanatory variables pertaining to Bourdieu's theory of sociocultural reproduction (parental education and the index of material possessions) are important predictors of educational attainment.

Therefore, this study confirms the research hypothesis based on the socio-cultural reproduction perspective (Bourdieu and Passeron 1977).

Moreover, it was identified that male students are less likely to complete upper-secondary education than female ones. Besides, no statistically significant difference exists between indigenous and non-indigenous students in the chances of completing this educational level after taking into account the effect of grades, access to scholarships and risk events. It can be posited that good academic performance and financial support during studies enable indigenous and non-indigenous students to move up through the different levels of upper-secondary. This finding might be explained due to the proportion of indigenous students identified in the sample, which is very small—only 4.8%-, are resilient students who have developed key skills and attitudes that allow them to move ahead in the curriculum. An alternative interpretation of this finding is that indigenous who complete

upper-secondary education are those who have moved from rural to urban areas and as a consequence, they have available a broader offer of educational options and resources.

The opportunities of completing upper-secondary education are also influenced by institutional factors such as the type of upper-secondary institution attended. More specifically, people who attended academic and private schools were more likely to complete upper-secondary education rather than drop out, whereas ones who studied in less selective institutions (technical schools and open and distance-learning systems) were less likely to complete upper-secondary education than those who attended more selective institutions. This finding is consistent with data published by the INEE (2014) which show that technical upper-secondary schools have constantly had higher dropout rates than general academic schools.

Also, no statistical difference was found between the chances of completing upper-secondary education of people who attended technological schools funded by state governments (Spanish abbreviations: Cecyte and Cecyt) and the chances of people who studied in general academic schools completing the same cycle. Conversely, students who attended federal technological schools were less likely to graduate from upper-secondary education than those who attended general academic institutions. This finding suggests that the Cecytes and Cecyts are performing better in terms of promoting upward educational mobility than other types of vocational institutions. Further studies of the curricular characteristics and practices of such institutions are therefore recommended.

Another important result is that variables related to the capability approach, such as freedom to choose school and people's engagement with/disengagement from school influence school completion. The weaker the student's engagement with school, the more likely s/he is to drop out. It bears stressing that the main variables used to measure school engagement/disengagement include the attribution of scant relevance to education, a preference for entering the labour market rather than studying, lack of interest in studies, disapproval of the school's disciplinary regulations, and the desire to transfer to a different type of institution. These results may be partially explained in terms of young people not valuing the upper-secondary curriculum offered to them because they find it irrelevant to their life aspirations.

Another interesting finding of this research is that a human agency dimension—freedom to choose school—matters for school completion. Those persons who reported that their school was chosen by their parents or assigned by the authorities were less likely to complete upper-secondary education than those who had a certain degree of freedom to choose their own school. This evidence confirms the research hypothesis grounded in the capability approach to the effect that Individuals' ability to move up through the different levels of upper-secondary and tertiary education is influenced by human-agency factors such as the degree of freedom/power to make educational choices. Persons who have more freedom/power to make educational decisions, such as choosing the type of institution they will attend, are more likely to move up through the education system.

Apart from that, access to scholarships in upper-secondary education is one of the strongest predictors of people undertaking and completing the said education. Individuals who received financial support were more likely to remain in school and complete their upper-secondary education than ones who did not receive such support. The financial support available to students included scholarships offered by the Ministry of Education, by the *Human Development Programme Opportunities, HDPO* (recently known in Spanish as Prospera programme), by the federal government, by the different Mexican states, and by universities.

It is noteworthy that the HDPO (Prospera programme) scholarships help some students to continue studying by addressing their financial constraints. The aforesaid Prospera

programme, which is the main policy tool designed to alleviate poverty in Mexico, focuses on enhancing the human capital of children and youths from poor families by investing in their education, health and nutrition. However, the School Dropout Survey does not specify any guidelines for granting scholarships from all the other funding sources. In other words, socioeconomic circumstances, academic achievement, or some other criterion, could have been considered when assigning the scholarships offered by the Ministry of Education, the federal government, the different Mexican states, and some universities.

Moreover, certain life events occurring during studies, such as parental problems, lack of safety on the way to school, sickness or death of a relative, family preferences for female education, marriage and pregnancy, increase the risk of dropping out of school. In other words, the more risk events a student faces in his/her life, the more likely s/he is to drop out of school.

Finally, higher levels of prior academic achievement (grades in lower-secondary education) were strongly associated with a higher likelihood of completing upper-secondary schooling. However, it should be borne in mind that, although grades are an important indicator of academic performance, this variable does not specify the extent to which people have developed other skills and abilities, such as mathematical, language, scientific, civic and citizenship competencies, among others.

Conclusion

This study has provided evidence that not only structural factors, such as sociocultural and economic resources, influence the attainment of higher levels of education but also human agency dimensions. The hypothesis that human agency—measured by individuals' degree of freedom to choose school—increases the likelihood of making higher educational transitions was corroborated. More precisely, the evidence indicates that people who reported having their school assigned by the authorities or chosen by their parents were less likely to complete upper-secondary education than those who said they had free choice. However, while freedom to choose in one educational stage, such as completion of upper-secondary education, is instrumentally relevant for completing that specific stage, it does not assure future benefits at subsequent educational phases (e.g., in higher education).

Consequently, it can be assumed that having a certain degree of freedom to make educational choices at one school transition does not necessarily mean that people will be free to make other educational decisions—e.g., to choose the university they will attend and the subjects they will study—in subsequent phases of their educational trajectories. In subsequent educational phases, such freedom of choice may be constrained not only by structural factors but also by institutional arrangements such as admission procedures. Unfortunately, there were no data available in large-scale data sets to explore the effects of human agency' dimensions when making the transition from upper-secondary to higher education in Mexico.

It also should be stressed that, since the variable, "freedom/power to participate in decision-making," is just one dimension of human agency, more research into other dimensions and sub-dimensions of agency needs to be undertaken in order to gain a deeper understanding of its role on persons' educational opportunities and outcomes.

Furthermore, recent educational policies in Mexico proposes some strategies for broadening participation in upper secondary education. One of these strategies explicitly aims to develop new models of open and distance-learning education (SEP 2013, 55). However, the evidence of this study suggest that students attending schools that have a certain component of distance-learning education are much less likely to complete upper secondary education

than those who studied in other types of schools: so, what justifies the expansion of a type of education that provides little benefits for individuals' educational trajectories?

Open and distance-learning educational programmes should be more attentive to the characteristics of the persons who attend this type of education. These schools seem to work under the assumption that individuals have a strong capacity for being self-directed or independent learners which might not necessarily be true. Individuals who enter at open and distance-learning programmes may be those who require more academic support.

To conclude, from a sociocultural-reproduction perspective, it would be advisable to broaden understandings of the role of structural influences related to class, gender and ethnicity on individuals' educational trajectories. From a capability perspective it is important to recognize human diversity and to foster agency and valued capabilities because they are both intrinsically and instrumentally relevant for persons' educational and life opportunities.

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References

Abril, E., R. Roman, M. Cubillas, and I. Moreno. 2008. "¿Deserción o autoexclusión? Un análisis de las causas de abandono escolar en estudiantes de educación media superior en Sonora, México." *Revista Electrónica de Investigación Educativa* 10 (1).

Bassi, M., M. Buso, and J. Muñoz. 2014. *Enrolment, Graduation, and Dropout Rates in Latin America. (Documento de Trabajo 170)*. Argentina: Universidad Nacional de la Plata.

Blaikie, N. 2010. *Designing Social Research*. 2nd ed. Cambridge: Polity Press.

Blanco, E. 2014. "Interrupción de la asistencia escolar: desigualdad social, instituciones y curso de vida." In *Caminos desiguales. Trayectorias educativas y laborales de los jóvenes en la Ciudad de México*, edited by E. Blanco, P. Solís, and H. Robles, 39–70. Mexico City: INEE-El Colegio de México.

Bourdieu, P. 1997. "The Forms of Capital." In *Education: Culture, Economy and Society*, edited by A. Halsey, H. Lauder, and A. Stuart Wells, 46–58. New York: Oxford University Press.

Bourdieu, P., and J.-C. Passeron. 1977. *Reproduction in Education, Society and Culture*. London: SAGE.

Breen, R., and J. Jonsson. 2005. "Inequality of Opportunity in Comparative Perspective: Recent Research on Educational Attainment and Social Mobility." *Annual Review of Sociology* 31: 223–243.

Brekke, I. 2014. "Long-term Labour Market Consequences of Dropping Out of Upper Secondary School: Minority Disadvantages?" *Acta Sociológica* 57 (1): 25–39.

Brighouse, H., and E. Untherhalter. 2010. "Education for Primary Goods or for Capabilities?" In *Measuring Justice: Primary Goods and Capabilities*, edited by H. Brighouse and I. Robeyns, 2662–2955. Cambridge: Cambridge University Press.

Flores-Crespo, P. 2002. "En busca de nuevas explicaciones sobre la relación entre educación y desigualdad. El caso de la Universidad Tecnológica de Nezahualcóyotl." *Revista Mexicana de Investigación Educativa* 7 (16).

Hart, C. 2013. *Aspirations, Education and Social Justice: Applying Sen and Bourdieu* (Kindle Ed.). London: Bloomsbury.

Henry, K. L., K. E. Knight, and T. P. Thornberry. 2012. "School Disengagement as a Predictor of Dropout, Delinquency and Problem Substance Use During Adolescence and Early Adulthood." *Journal of Youth and Adolescence* 41: 156–166.

Ibrahim, S., and S. Alkire. 2007. "Agency and Empowerment: A Proposal for Internationally Comparable Indicators." *Oxford Development Studies* 35 (4): 379–403.

INEE. 2014. *Panorama Educativo de México. Indicadores del Sistema Educativo Nacional 2013. Educación Básica y Media Superior*. México: INEE.

INEE. 2018. *Panorama Educativo de México 2017. Indicadores del Sistema Educativo Nacional. Educación básica y media superior*. México: INEE.

Kerckhoff, A. 1993. *Diverging Pathways: Social Structures and Career Deflections*. New York: Cambridge University Press.

Lamb, S., and E. Markussen. 2011. "School Dropout and Completion: An International Perspective." In *School Dropout and Completion: International Comparative Studies in Theory and Policy*, edited by S. Lamb, E. Markussen, R. Teese, N. Sandberg, and J. Polesel, 1–20. Dordrecht: Springer.

Long, S., and J. Freese. 2014. *Regression Models for Categorical Dependent Variables Using Stata*. College Station, TX: Stata Press. Edición de Kindle.

Lucas, S., and L. Beresford. 2010. "Naming and Classifying: Theory, Evidence and Equity in Education." *Review of Research in Education* 34: 25–84.

Roderick, M. 1994. "Grade Retention and School Dropout: Investigating the Association." *American Educational Research Journal* 31 (4): 729–759. doi:10.3102/00028312031004729.

Rowlands, J. 1997. *Questioning Empowerment: Working with Women in Honduras*. Oxford: Oxfam.

Rumberger, R. W. 1995. "Dropping Out of Middle School: A Multilevel Analysis of Students and Schools." *American Educational Research Journal* 32 (3): 583–625. doi:10.3102/00028312032003583.

Rumberger, R. 2011. "High School Dropouts in the United States." In *School Dropout and Completion: International Comparative Studies in Theory and Policy*, edited by S. Lamb, E. Markussen, R. Teese, N. Sandberg, and J. Polesel, 275–294. Dordrecht: Springer.

Rumberger, R. W., and S. L. Thomas. 2000. "The Distribution of Dropout and Turnover Rates Among Urban and Suburban High Schools." *Sociology of Education* 73: 39–67. doi:10.2307/2673198.

Secretaria de Educación Pública (SEP). 2013. *Programa Sectorial de Educación 2013-2018*. México: SEP.

Sen, A. 1985. *Commodities and Capabilities*. Amsterdam: Elsevier.

Sen, A. 1992. *Inequality Re-examined*. Oxford: Oxford University Press.

Sen, A. 1999. *Development as Freedom* (Kindle Ed.). Oxford: Oxford University Press.

SEP-SEMS. 2012. *Reporte de la encuesta nacional de deserción de la educación media superior*. Mexico: SEP-CENEVAL.

Shavit, Y., and W. Muller. 2000. "Vocational Secondary Education" *European Societies* 2 (1): 29–50.

Székely, M. 2010. "Avances y transformaciones en la educación media superior." In *Los grandes problemas de México*. Vol. 7., edited by A. Arnaut and S. Giourguli, 313–336. Mexico: El Colegio de México.

Thornberry, T. P., M. Moore, and R. L. Christenson. 1985. "The Effect of Dropping Out of High School on Subsequent Criminal Behavior." *Criminology* 23 (1): 3–18.

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